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Evaluate yourself :

1. (i) **Element** - Copper wire, Silver plate

(ii) **Compound** - Sugar, distilled water, carbon dioxide, Table salt, Naphthalene balls

(iii) **Mixture** - Sea water

2. (i) **C_2H_5OH** : (2 x 12) + (5 x 1) + (1 x 16) + (1 x 1) = **46 g**

(ii) **$KMnO_4$** : (1 x 39) + (1 x 55) + (4 x 16) = **158 g**

(iii) **$K_2Cr_2O_7$** : (2 x 39) + (2 x 52) + (7 x 16) = **294 g**

(iv) **$C_{12}H_{22}O_{11}$** : (12 x 12) + (22 x 1) + (11 x 16) = **342 g**

3. (a) Molar mass of ethane, C_2H_6 = (2 x 12)+ (6 x 1) = 30 g mol^{-1}

n = mass / molar mass = 9 g / 30 g mol^{-1} = **0.3 mole**

(b) At 273 K and 1 atm pressure 1 mole of a gas occupies a volume of 22.4 L

Therefore,

number of moles of oxygen, that occupies a volume of 224 ml at 273 K and 3 atm pressure

$$= \frac{1\,\text{mole}}{273\,\text{K} \times 1\,\text{atm} \times 22.4\,\text{L}} \times 0.224\,\text{L} \times 273\,\text{K} \times 3\,\text{atm}$$

$$= 0.03\,\text{mole}$$

1 mole of oxygen contains 6.022 x 10^{23} molecules

0.03 mole of oxygen contains = 6.022 x 10^{23} x 0.03

= **1.807 x 10^{22} molecules of oxygen**

4. (a) Mass of the metal = 0.456 g

Mass of the metal chloride = 0.606 g

0.456 g of the metal combines with 0.15 g of chlorine.

Mass of the metal that combines with 35.5 g of chlorine is $\frac{0.456}{0.15} \times 35.5$

= 107.92 g eq^{-1}.